SHORT-LIFETIME H_0 PERSISTENCE FOR THE THREE-DIMENSIONAL SIDE-24 PERIODIZED BARGMANN–FOCK FIELD

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ABSTRACT. Let f_{24} be the normalized periodization to the cubic torus $\mathbb{T}^3_{24}$ of the real Bargmann–Fock field with covariance $\exp(-|x-y|^2/2)$. We consider finite, nonessential superlevel-set H_0 persistence bars and their first-moment lifetime density per unit volume. We prove the short-lifetime law

$$\nu_{3,24}(\ell) = c_{3,24}\,\ell^{-1/3}\big(1+O(\ell^{1/3})\big), \qquad \ell \downarrow 0,$$

where $c_{3,24} > 0$ is given by an explicit finite-dimensional Gaussian-cone integral. The proof identifies the dimension-cancelling local fold mechanism, proves that, for compact positive marks, elder-rule selection changes the all-typed local contact count by only $O(r^3)$ in Palm probability, and fixes all ordered-pair, directed-sphere, Kac–Rice and lifetime-Jacobian normalizations. The coefficient is related to its value for the Bargmann–Fock field on $\mathbb{R}^3$ by

$$\frac{c_{3,24}}{c_{3,\infty}} - 1 = -\frac{620813376}{35}e^{-288} + \varepsilon_{24}, \qquad |\varepsilon_{24}| < 10^{-216},$$

with $c_{3,\infty} = 2^{-23/6}3^{-8/3}(29\sqrt{6}-36)\,\Gamma(1/6)\,\pi^{-5/2}$. We also state the three-term expansion $\nu_{3,24}(\ell) = c_{3,24}\ell^{-1/3} + c_1\ell^{1/4} + c_2\ell^{1/3} + O(\ell^{3/7})$, with explicit constants $c_1 < 0$ and c_2, and describe the cusp and fold mechanisms that produce the corrections.

STATUS OF THIS DRAFT (REMOVE BEFORE SUBMISSION)

What this is. An author-side V3 draft, written by an AI assistant under the author's direction: Anthropic Claude, Claude Code session `session_01NMeKEismAyeqgdB4sy2NJU`. It was assembled on 13 October 2026 and revised through r15 on 21 October 2026. It integrates the frozen V1 text (30 July 2026; source SHA-256 `fb576de6...`), the V2 amendments M1–M7 (1 August 2026) and the *apply* items of the V3 edit pack (E1–E78, status of 13 October 2026). No mathematical statement here is new: every result is the manuscript's own or is taken, with its stated scope, from the repository records listed in Appendix A. The companion file `V3_CHANGE_LIST.md` gives the placement of every amendment and edit-pack item, and the items deliberately not printed.

Author decisions (decided). The author kept the drafted defaults of decisions 1–3 and chose the Supplement-only form of decision 4 (main#389, comment 6324313740).

(1) *Strength of the main statement* (edit pack, decision 1). Theorem 2.1 states the two-provider-reviewed form $c\ell^{-1/3}(1+O(\ell^{1/3}))$, proved in §3. The stronger three-term law is Theorem 2.3, with its proof in the supplementary records; the later refinements are Remark 2.4.
(2) *Human review* (decision 2). Worded as a disclosure (§7): no independent human specialist review, following the author's statement that none is planned. The draft says nothing about the author's own reading; add a sentence there if you wish.
(3) *Planar material* (decision 3). Results for $d = 2$ appear only as Discussion remarks (§5.5); the theorem is $d = 3$.

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2020 *Mathematics Subject Classification.* Primary 60G60; Secondary 60G15, 55N31, 60D05.
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(4) *Relation to the parent record* (decision 4, from a same-provider referee read of r3). Decided: the leading term rests on the Supplement alone. The proofs of Corollary 3.5 and Proposition 3.10 cite Supplement Theorem S12.3 (Theorem A of [R2], with its proof in Supplement Part III) and Supplement Proposition S6.2. §6 states which results are cases of the parent record and which are specific to the side-24 field, and Supplement Part III (Sections S12–S13) states and proves the parent record's results that the Supplement uses.

Review status of each stated result (internal tags: R·2P = nonauthor reviews from two providers; R·1P = one provider; +GB = an additional nonauthor read by Grok Bot support agents (Cursor-hosted, not the xAI model lane); READ = read in bounded nonauthor slices; all reviews are AI reviews posted from one account, so no organizational independence is claimed).

Statement	Status	Records
Thm. 2.1, leading term	V1/V2 proof (§3, Supplement Part I); parent chain with provider-distinct AI reviews, transcribed in Supplement Part III; every section of the Supplement READ (+GB), not yet read as a whole	[R1], [R2]
Thm. 2.1, rate $O(\ell^{1/3})$	R·2P (xAI R1–R6; Anthropic full-depth read), merged	[R3]
(2.10), $\lvert\varepsilon_{24}\rvert < 10^{-216}$	R·2P, merged	[R4]
Cor. 3.5, upper bound (3.13)	Theorem A of the parent chain, accepted at existential scope; nonauthor reviews by xAI and Anthropic, every interface read at full depth by at least one. The second route in Supplement §S5 has provider-distinct reads: by xAI of §§S5.1–S5.3 and S5.5–S5.7 (PASS_SCOPED), and by OpenAI of §S5.4, which found a gap in its finite-scale witness densities; Supplement draft 9 repairs it (§S5.8), and the repair has nonauthor reads by OpenAI (PASS_SCOPED, bound to the TEX source) and by a Grok Bot support agent (PASS)	[R2], [R1]
Cor. 3.5, lower bound	R·1P (OpenAI), merged; constants existential	[R6]
Rem. 3.7	merged with OpenAI reads and an Anthropic composition read (R·2P), author-side at their scope	[R10]
Rem. 3.11	R·2P at the contact-integral scope	[R20]
Thm. 2.3	the links in [R11], and c_1 and the closed form of c_2, each merged with an OpenAI/Codex nonauthor review and a Grok Bot read (R·1P+GB), and certified at those stated scopes; the torus transfer of c_2 has an OpenAI `ACCEPT_SCOPED` of slices A–C and an Anthropic acceptance of slices A and C, and no Grok Bot read; the identification $c_2 = c_2[K_{24}]$ is Supplement Corollary S11.11, with a Grok Bot read of its proof in draft 10 (PASS after one AMEND) and an OpenAI read of its proof at draft 12 (`PASS_SCOPED`, 20 October 2026)	[R11], [R12]
Rem. 2.4	READ (same account), all records on `Math-main`; the value in (c) at the conditional scope of [R15, Thm. A]	[R14]
Rem. 2.5	Theorem Z R·2P, and proved in the Supplement (§S8.6, draft 12); Theorems U, K, P R·1P, with the bound (K2) of Theorem K proved in the Supplement (Lemma S8.16, draft 11); the $c_3\ell^{2/3}$ term and $\delta = 1/50$ READ; $B_{3,24} \approx 10.2$ exploration only	[R7], [R13], [R14], [R8]
§5	labelled in place: proved, read, author-side, numerical, or expectation	App. A

Not established, and not claimed anywhere below. An explicit ℓ-range on which the relative error in (2.1) is below a given δ (all remainder constants are existential); a theorem for side lengths $L \neq 24$ beyond the identities cited, or for the interchange of $L \to \infty$ and $\ell \downarrow 0$; a

numerical value of ε_{24}; a directed-rounding certificate for the integral (2.9) beyond the relative bound $|c_{3,24}/c_{3,\infty} - 1| < 10^{-106}$ [R5]; any statement for H_k, $k > 0$.

Before submission. Remove this section; have the Supplement [R1] read as a whole; run the MathSciNet/zbMATH pass (§6); choose the venue and do its style pass.

1. Field, filtration, and lifetime intensity

Write $\mathbb{T}^3_{24} = (\mathbb{R}/24\mathbb{Z})^3$. The normalized periodized Bargmann–Fock covariance is

$$K_{24}(z) = Z_{24}^{-1} \sum_{n\in\mathbb{Z}^3} \exp\Big(-\frac{|z+24n|^2}{2}\Big), \qquad Z_{24} = \sum_{n\in\mathbb{Z}^3} e^{-288|n|^2}. \tag{1.1}$$

Let f_{24} be a centered stationary Gaussian field with covariance K_{24}.

Proposition 1.1 (genericity of the side-24 field)**.** *The field f_{24} has real-analytic sample paths and, with probability one, is Morse and has pairwise distinct critical values.*

Proof. In the Fourier basis $e_m(z) = \exp(2\pi i\, m\cdot z/24)$, Poisson summation gives

$$\widehat{K}_{24}(m) = \frac{(2\pi)^{3/2}}{24^3 Z_{24}} \exp\Big(-\frac{2\pi^2|m|^2}{24^2}\Big) > 0, \qquad m\in\mathbb{Z}^3. \tag{1.2}$$

Write the sample series using standard Gaussian Fourier coefficients with amplitudes $\widehat{K}_{24}(m)^{1/2}$. For every $R > 0$,

$$\sum_{m\in\mathbb{Z}^3} \mathbb{E}|\xi_m|\, \widehat{K}_{24}(m)^{1/2} \exp(2\pi R|m|/24) < \infty.$$

Fubini therefore gives almost-sure absolute and locally uniform convergence on every complex strip $\{z\in\mathbb{C}^3 : |\operatorname{Im} z| \le R\}$. The sample series has an entire periodic extension, so its restriction to the real torus is real analytic.

If T is a finite linear combination of derivative evaluations, then

$$\operatorname{Var}(Tf_{24}) = \sum_{m\in\mathbb{Z}^3} \widehat{K}_{24}(m)\, |\widehat{T}(m)|^2. \tag{1.3}$$

Because every $\widehat{K}_{24}(m)$ is positive and Fourier coefficients determine distributions on the torus, this variance vanishes only when the corresponding finite derivative distribution is zero. At one point, the derivatives of the Dirac distribution indexed by distinct multi-indices are linearly independent. Hence the gradient and symmetric-Hessian coordinate vector is nondegenerate at every point. The smoothness, density and modulus conditions of [2, Theorem 11.3.4] follow from the analytic covariance, this nondegeneracy and compactness, and that theorem implies that f_{24} is Morse almost surely.

For distinct critical values, fix $\delta > 0$ and consider the compact six-dimensional configuration set

$$\mathcal{T}_\delta = \{(x,y)\in(\mathbb{T}^3_{24})^2 :\ \operatorname{dist}(x,y)\ge\delta\}. \tag{1.4}$$

Define the seven-dimensional Gaussian field

$$\mathcal{H}(x,y) = (\nabla f_{24}(x), \nabla f_{24}(y), f_{24}(x) - f_{24}(y)). \tag{1.5}$$

For $x \ne y$, the derivative distributions in (1.5) are linearly independent: distributions supported at distinct points have disjoint supports, and the coefficient of $\delta_x - \delta_y$ must vanish separately from all derivative coefficients. Equation (1.3) therefore makes the covariance of $\mathcal{H}(x,y)$ positive definite. On $\mathcal{T}_\delta$, compactness gives a uniform covariance eigenfloor and hence uniformly bounded one-point densities. Apply [2, Lemma 11.2.10] in each member of a finite coordinate cover of $\mathcal{T}_\delta$. Since $\mathcal{H}$ is a C^1 field from a six-dimensional compact set to $\mathbb{R}^7$,

$$\mathbb{P}\{\mathcal{H}^{-1}(0)\cap\mathcal{T}_\delta \ne \varnothing\} = 0. \tag{1.6}$$

Taking the countable union over $\delta = 1/n$ shows that no two distinct critical points have the same value. □

For the superlevel filtration

$$X_a = \{x \in \mathbb{T}^3_{24} : \ f_{24}(x) \geq a\}, \tag{1.7}$$

each non-global local maximum creates an H_0 class and each finite class dies at a unique index-two merging saddle. If M is the creating maximum and S its death saddle, its lifetime is

$$\ell = f_{24}(M) - f_{24}(S) > 0. \tag{1.8}$$

Proposition 1.2 (existence of the first-moment lifetime density)**.** *Let M_* be the global maximum and let $D_f(M)$ be the elder-rule death saddle of a non-global maximum M. The finite measure*

$$\Lambda_{3,24}(A) = 24^{-3}\,\mathbb{E} \sum_{M \neq M_*} \mathbf{1}_{\{f_{24}(M) - f_{24}(D_f(M)) \in A\}}, \qquad A \subset (0,\infty), \tag{1.9}$$

is absolutely continuous with respect to Lebesgue measure.

Proof. Proposition 1.1 makes the elder pairing unambiguous almost surely. A Morse function on the compact torus has finitely many critical points, and the one-point Kac–Rice formula gives a finite expected critical-point count. Hence (1.9) is a finite measure.

For $\delta > 0$, restrict the ordered maximum–index-two-saddle pair process to $\operatorname{dist}(M, S) \geq \delta$. The elder-pair indicator is a bounded measurable mark of the finite Morse filtration (Lemma 3.2). On that compact configuration set, complete positive Fourier support makes

$$(f_{24}(x), f_{24}(y), \nabla f_{24}(x), \nabla f_{24}(y))$$

nondegenerate, uniformly on that set. The two-point Kac–Rice formula, with the bounded measurable mark that the ordered pair is the elder pair, therefore expresses the expected birth–death measure as an integral against $db\,dh$. Thus the restricted measure is absolutely continuous on $\{(b,h) : b > h\}$.

Every distinct critical pair belongs to one of the sets $\{\operatorname{dist}(M,S) \geq 1/n\}$. Monotone convergence over this countable cover shows that the full expected birth–death measure is absolutely continuous. Finally, if $A \subset \mathbb{R}$ has one-dimensional Lebesgue measure zero, then

$$\{(b,h) : \ b - h \in A\} \tag{1.10}$$

has two-dimensional Lebesgue measure zero by Fubini after the linear change of variables $(b,h) \mapsto (b, b-h)$. The lifetime pushforward is therefore absolutely continuous. □

Let $\nu_{3,24}(\ell)\,d\ell = d\Lambda_{3,24}(\ell)$. In the proof below, $\nu_{3,24}$ denotes the pointwise Kac–Rice representative of Convention 2.2.

2. Main results

Theorem 2.1 (fixed-side short-lifetime law)**.** *For the normalized side-24 field (1.1), the finite nonessential superlevel H_0 first-moment lifetime density satisfies*

$$\nu_{3,24}(\ell) = c_{3,24}\,\ell^{-1/3}(1 + O(\ell^{1/3})), \qquad \ell \downarrow 0, \tag{2.1}$$

equivalently $\nu_{3,24}(\ell) = c_{3,24}\ell^{-1/3} + O(1)$, where $0 < c_{3,24} < \infty$ is given by (2.9) below.

Fix $t \in S^2$ and an orthonormal transverse frame n_1, n_2. Define

$$u_t = -\frac{\partial_t^3 f_{24}}{12}, \qquad g_t = (\partial_t f_{24}, \partial_{n_1} f_{24}, \partial_{n_2} f_{24}), \tag{2.2}$$

$$h_t = (\partial_{tt} f_{24}, \partial_{tn_1} f_{24}, \partial_{tn_2} f_{24}), \tag{2.3}$$

and

$$q_t = (\partial_{n_1 n_1} f_{24}, \partial_{n_1 n_2} f_{24}, \partial_{n_2 n_2} f_{24}), \qquad Q_t(q) = \begin{pmatrix} q_1 & q_2 \\ q_2 & q_3 \end{pmatrix}. \tag{2.4}$$

Put

$$\Sigma_g(t) = \mathrm{Cov}(g_t), \qquad \Sigma_h(t) = \mathrm{Cov}(h_t), \tag{2.5}$$

$$\sigma_t^2 = \mathrm{Var}(u_t) - \mathrm{Cov}(u_t, g_t)\, \Sigma_g(t)^{-1}\, \mathrm{Cov}(g_t, u_t), \tag{2.6}$$

$$\Gamma_t = \mathrm{Cov}(q_t) - \mathrm{Cov}(q_t, h_t)\, \Sigma_h(t)^{-1}\, \mathrm{Cov}(h_t, q_t), \tag{2.7}$$

and

$$D_t = \mathbb{E}\Big[(\det Q_t(q_t))^2\, \mathbf{1}_{\{Q_t(q_t)<0\}} \;\Big|\; h_t = 0\Big]. \tag{2.8}$$

Then

$$c_{3,24} = B_3 \int_{S^2} \frac{(\sigma_t^2)^{2/3} D_t}{\sqrt{\det \Sigma_g(t)\, \det \Sigma_h(t)}}\, dt, \qquad B_3 = \frac{3\,\Gamma(7/6)}{2^{4/3}\pi^{7/2}}. \tag{2.9}$$

The formula is independent of the transverse-frame choice.

Moreover,

$$\frac{c_{3,24}}{c_{3,\infty}} - 1 = -\frac{620813376}{35}\, e^{-288} + \varepsilon_{24}, \qquad |\varepsilon_{24}| < 10^{-216}, \tag{2.10}$$

where $c_{3,\infty}$, the value of (2.9) for the Bargmann–Fock field on $\mathbb{R}^3$, is

$$\begin{aligned} c_{3,\infty} &= 2^{-23/6} 3^{-8/3} (29\sqrt{6} - 36)\, \Gamma(1/6)\, \pi^{-5/2} \\ &= 0.041775931840598343342936665428575556466681519 6\ldots. \end{aligned} \tag{2.11}$$

In particular $c_{3,24} < c_{3,\infty}$:

$$\frac{c_{3,24}}{c_{3,\infty}} - 1 = -1.486217029065827607 22\ldots \cdot 10^{-118} + \varepsilon_{24},$$

so the displayed correction is the exact leading term to a relative precision better than 10^{-98}.

The leading term of (2.1), with the formula (2.9), is proved in §§3.1–3.6; the value (2.11) in §3.7; the correction (2.10) in §3.8; and the rate $O(\ell^{1/3})$ in §3.9.

Convention 2.2 (density)**.** Throughout, "density" means the ℓ-density of the expected counting measure of finite nonessential superlevel H_0 bars per unit volume, with the Kac–Rice counting measure normalized as in Proposition 1.2 and §§3.1–3.3 (marked two-point formula, ordered typed pairs, per-volume normalization $24^{-3} \int dx = 1$). Pointwise statements about $\nu_{3,24}$ and about the candidate and rejected densities of Remark 2.5, here and below, refer to their canonical marked Kac–Rice density versions: the representatives given by the marked two-point formula, written in the pair coordinates of §3.1 for separations below a fixed $\varrho \in (0, 12)$ and in the coordinates (M, S) for the larger separations (Supplement (S6.7) and (S6.10) for $\nu_{3,24}$, and the same formulas without the elder mark for the candidate density). The substitution $\kappa = 6\ell/r^3$ shows that their sum does not depend on ϱ. A change of density on a null set need not preserve a pointwise asymptotic. All $o(\cdot)$ and $O(\cdot)$ statements are as $\ell \downarrow 0$, uniform on compact positive mark sets; the statements about the densities themselves carry no marks.

2.1. **Beyond the leading term.** The rate in (2.1) is not sharp. Theorem 2.3 is proved in the records listed in Appendix A; Part II of the Supplement states the results it uses, with proofs given in full or condensed. Remarks 2.4 and 2.5 report further results of those records, at the review status and scope stated with each. None of them is used in the proof of Theorem 2.1; their review status is described in §7.

Theorem 2.3 (three-term law)**.** *For the canonical marked Kac–Rice density version of* $\nu_{3,24}$*,*

$$\nu_{3,24}(\ell) = c_{3,24}\,\ell^{-1/3} + c_1\,\ell^{1/4} + c_2\,\ell^{1/3} + O(\ell^{3/7}), \qquad \ell \downarrow 0, \tag{2.12}$$

where the constants are as follows.

(i) $c_1 < 0$ *is the cusp coefficient*

$$c_1 = -\frac{192}{7}\,2^{1/4}\int_{S^2}\int_{\mathbb{R}} \pi_0(u; v_0(b,0))\,\mathbb{E}_0\Big[|Y|^{7/4}\,|\det A|^{1/4}\,\mathbf{1}_{\{A<0\}}\;\Big|\; b\Big]\,db\,d\sigma(u), \tag{2.13}$$

where, at a zero-gap contact in direction u*,* $A = D_\perp^2 f$ *is the transverse Hessian,* $\gamma = \nabla_\perp \partial_u^2 f$*,* $f_4 = \partial_u^4 f$ *and* $Y = \frac{f_4}{12}\det A - \frac{1}{4}\gamma^\top \operatorname{adj}(A)\gamma$*, and* π_0*,* $\mathbb{E}_0$ *are the zero-gap pin density and conditional expectation of [R11] (Theorem CU).*

(ii) c_2 *is the fold-scale finite part of the* r^2*-coefficient of the two-point kernel. At fixed birth height* b*, gap variable* k *(the gap is* kr^3*, so* $k = \kappa/6$ *in the notation of (3.5)) and direction* u*, the two-point kernel* $\mathcal{A}_r(b,k,u)$ *of [R11] has the fold expansion* $\mathcal{A}_r = \mathcal{A}_0 + r^2\mathcal{A}_2 + O(r^3(1+k^{-1}))$*, and*

$$c_2 = \frac{1}{3}\int_{S^2}\int_{\mathbb{R}}\int_0^\infty \left[\mathcal{A}_2(b,k,u) - \mathcal{A}_2(b,0,u)\right] k^{-4/3}\,dk\,db\,d\sigma(u),$$

the Hadamard finite part at $k = 0$ *of* $\frac{1}{3}\int \mathcal{A}_2\,k^{-4/3}$ *[R11]. In the notation of §3, the kernel is* $\mathcal{A}_r(b,\kappa/6,t) = G_r^{\mathrm{all}}(t,b,\kappa)$*, by Supplement §S6.*

(iii) Enclosures at $L = 24$*. First,* $c_1/c_{3,24} \in [-5.07106215,\ -5.07106213]$*, a certified enclosure of the integral (2.13) divided by* $c_{3,24}$*, given the enclosure* $0.04177593184059834334 < c_{3,24} < 0.04177593184059834335$ *of [R5] [R12]; Supplement Corollary S9.16 is that enclosure, and its reading as the coefficient of* $\ell^{7/12}$ *in (2.14) uses Theorem CU (Supplement Theorem S9.2 and Remark S9.17). Second, the fixed-cone coefficient expression* $c_2[K_{\mathrm{ref}}]$ *of the Gaussian kernel* $K_{\mathrm{ref}}(z) = e^{-|z|^2/2}$ *on* $\mathbb{R}^3$ *(Supplement Definition S11.2; [R12]) has a closed form with a certified enclosure,*

$$c_2[K_{\mathrm{ref}}] \in [0.1612340491269447,\ 0.1612340491269810]$$

(Supplement Theorem S11.5). The same expression for the side-24 covariance, $c_2[K_{24}]$*, differs from it by less than* 10^{-60} *(Supplement Corollary S11.10), and* $c_2 = c_2[K_{24}]$*: Supplement Corollary S11.11 derives from the fold expansion of Supplement §S10 the two identifications that [R12] states as interfaces. So* $c_2 \in [0.1612340491269446,\ 0.1612340491269811]$*. Third, with the enclosure of* $c_{3,24}$ *in [R5],*

$$c_2/c_{3,24} \in [3.859496174547456,\ 3.859496174548331]$$

(Supplement Corollary S11.12).

Equivalently,

$$\nu_{3,24}(\ell) = c_{3,24}\,\ell^{-1/3}\Big(1 + \tfrac{c_1}{c_{3,24}}\,\ell^{7/12} + \tfrac{c_2}{c_{3,24}}\,\ell^{2/3} + O(\ell^{16/21})\Big), \qquad \tfrac{c_1}{c_{3,24}} \approx -5.0711, \quad \tfrac{c_2}{c_{3,24}} \approx 3.8595. \tag{2.14}$$

The proof combines Theorem R^+ (the bounded remainder of Theorem 2.1 vanishes for elder pairs), Theorem R^{++} (its rate $\ell^{1/4}$), Theorem CU (the cusp term), Theorem T and Theorems E3, $\mathrm{E3}^+$ (the fold-scale term and the remainder $\ell^{3/7}$) with the far elder bound of [R9]; see [R11]. The mechanism behind c_1 and c_2 is described in §5.2.

Remark 2.4 (the next order). The records [R14] carry the expansion one order further.

(a) The remainder in (2.14) is $O(\ell)$:

$$\nu_{3,24}(\ell) = c_{3,24}\,\ell^{-1/3}\Big(1 + \tfrac{c_1}{c_{3,24}}\ell^{7/12} + \tfrac{c_2}{c_{3,24}}\ell^{2/3} + O(\ell)\Big). \tag{2.15}$$

In particular the elder density has no $\ell^{1/2}$ term.

(b) More precisely,

$$\nu_{3,24}(\ell) = c_{3,24}\,\ell^{-1/3} + c_1\ell^{1/4} + c_2\ell^{1/3} + (c_3 - R_{2/3})\,\ell^{2/3} + o(\ell^{2/3}), \tag{2.16}$$

where c_3 is the $\ell^{2/3}$ coefficient of the candidate density (Remark 2.5) and $R_{2/3}$ that of the rejected density; both are explicit integrals.

(c) At $L = 24$, and at the conditional scope of [R15, Theorem A],

$$(c_3 - R_{2/3})(24) \in [0.0773381,\, 0.0775860], \qquad \text{i.e.} \quad 0.07746 \pm 0.00013 \approx 1.854\,c_{3,24},$$

with $c_3(24) = 0.016092311 \pm 10^{-9}$ and $R_{2/3}(24) < 0$. These enclosures are computer-assisted, in standard-library interval arithmetic. So the coefficient of ℓ in the relative expansion is positive.

Remark 2.5 (candidate, elder and rejected densities). Call *candidate pairs* all ordered typed (maximum, index-two saddle) pairs at every separation, counted by the two-point Kac–Rice formula with the normalization of Convention 2.2, with density $\nu_{3,24}^{\mathrm{cand}}$, and let $\rho_{\mathrm{rej}} = \nu_{3,24}^{\mathrm{cand}} - \nu_{3,24}$ be the density of the candidate pairs that the elder rule rejects. Without the compact-mark restriction of Corollary 3.5, ρ_{rej} is not $o(1)$ (Theorem U of [R7]): it is bounded (Theorem K and Corollary T there) and converges, as $\ell \downarrow 0$, to a positive constant $B_{3,24}$, an explicit Kac–Rice integral of the equal-height unmarked two-point kernel (Theorem Z there). Consequently ρ_{rej} has inverse-moment threshold, the infimum of the exponents p with $\int_0^1 \ell^p \rho(\ell)\,d\ell < \infty$, exactly -1, whereas $\nu_{3,24}^{\mathrm{cand}}$ and $\nu_{3,24}$ keep the threshold $-2/3$ of the $\ell^{-1/3}$ law. Theorem 2.1 concerns the elder density and is unaffected; an earlier conjecture of an $O(\ell^{2/3})$ unrestricted difference is withdrawn [R7]. More precisely,

$$\nu_{3,24}^{\mathrm{cand}}(\ell) = c_{3,24}\,\ell^{-1/3} + B_{3,24} + I^{\mathrm{cand}}\ell^{1/4} + c_2\,\ell^{1/3} + c_3\,\ell^{2/3} + O(\ell^{2/3+\delta}), \qquad \delta = \tfrac{1}{50}, \tag{2.17}$$

with $I^{\mathrm{cand}} = \frac{3^{1/4}}{2}c_1$ [R11, R13]; the term $c_3\ell^{2/3}$ and the rate δ are from notes C3 and C3-R [R14]. In particular the candidate density has no $\ell^{1/2}$ term, by a reflection parity of the pinned two-point problem (Theorem P of [R13]). Subtracting (2.12) from (2.17),

$$\rho_{\mathrm{rej}}(\ell) = B_{3,24} + (1 - \tfrac{3^{1/4}}{2})|c_1|\,\ell^{1/4} + O(\ell^{3/7}), \tag{2.18}$$

so the rejected density approaches its limit from above, and the fold-scale coefficient c_2 is common to the candidate and elder densities; by Remark 2.4 the next term of ρ_{rej} is $R_{2/3}\ell^{2/3}$. Numerically, as exploration and not as an enclosure, $B_{3,24} \approx 10.2$, about $240\,c_{3,24}$: up to a near-field correction of about -0.02 it equals $24^3 \int \rho_{\max}(b)\rho_{\mathrm{sad}}(b)\,db$, the overlap of the height densities of maxima and index-two saddles [R8]. Unmarked small-gap pairs on the side-24 torus are thus overwhelmingly unrelated far-apart pairs, which the elder rule removes.

3. Proof of Theorem 2.1

The technical lemmas used below are proved in the Supplement [R1]. It assembles the exact technical appendix and its gap-fill supplement of July and August 2026, with the corrections recorded there. Statements quoted from it carry their Supplement numbers (S1.1, S1.2, ...).

3.1. **Ordered pair coordinates and unit multiplicity.** For a maximum M and an index-two saddle S at distance $r < 12$, let

$$x = \frac{M+S}{2}, \qquad S - M = rt, \qquad t \in S^2. \tag{3.1}$$

The inverse local Euclidean map is

$$(x, u) \longmapsto (M, S) = (x - u/2,\ x + u/2). \tag{3.2}$$

Its determinant has absolute value one, and therefore

$$dM\, dS = dx\, du = dx\, r^2 dr\, d\sigma_2(t). \tag{3.3}$$

The antipodal direction $-t$ exchanges M and S; it does not represent the same typed ordered pair. Thus the full directed sphere S^2 counts each maximum–saddle pair once, with no factor $1/2$.

For a Morse function with distinct critical values, each finite nonessential superlevel H_0 bar has one creating maximum and one merging death saddle. An index-two merge reduces β_0 by one, so the elder pairing has unit multiplicity. Counting critical-point pairs, rather than unstable-branch incidences, introduces no branch factor.

Palm law and the elder mark. Write $\Pi_{r,x,t} = (f(M), f(S), \nabla f(M), \nabla f(S))$ for the pins of the pair and $z_r(b, \kappa) = (b,\ b - \kappa r^3/6,\ 0,\ 0)$ for their target (cf. (3.5) below).

Definition 3.1 (Palm law and elder mark; Supplement Definitions S5.1 and S5.2)**.** The corrected-pin covariance is nondegenerate (by (1.3); Supplement Lemma S2.3(i)), so the regular conditional Gaussian law $\mathbb{E}^0 = \mathbb{E}[\,\cdot \mid \Pi_{r,x,t} = z_r(b,\kappa)]$ exists. With

$$W_r = |\det H_M|\,|\det H_S|\,\mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S)=2\}}, \qquad Z_r = \mathbb{E}^0[W_r] > 0 \tag{3.3a}$$

(Supplement Lemma S2.9(i)), the determinant-weighted Palm law is $\mathbb{E}_r^{MS}[F] = \mathbb{E}^0[W_r F]/Z_r$. For continuous g on $\mathbb{T}^3_{24}$, the *maximin connection level* of a point x is

$$d_g(x) = \sup\Big\{ \min_{s\in[0,1]} g(\gamma(s)) :\ \gamma \text{ a path with } \gamma(0) = x,\ g(\gamma(1)) > g(x) \Big\},$$

with $\sup \emptyset = -\infty$. The *elder mark* of a typed pair (M, S) with $f(S) < f(M)$ is $e(f; M, S) = \mathbf{1}\{d_f(M) = f(S)\}$.

Lemma 3.2 (Supplement Lemma S5.3 and Definition S5.4)**.** *The elder mark is Borel measurable. On the Morse locus, $e(f; M, S) = 1$ if and only if the two upper sectors of S lie in distinct components of $\{f > f(S)\}$ and M is the younger of the two birth maxima; this is the elder-pair condition of Proposition 1.2. With p_Π the Gaussian density of the pins $\Pi_{r,x,t}$ and the all-typed contact intensity $K_r^{\mathrm{all}} = p_\Pi(z_r(b,\kappa))\, Z_r$, the selected contact intensity is*

$$K_r^{\mathrm{sel}} = p_r\, K_r^{\mathrm{all}}, \qquad p_r = \mathbb{E}_r^{MS}[e]. \tag{3.3b}$$

3.2. **Contact scaling and the universal radial measure.** Put

$$M = x - \frac{r}{2}t, \qquad S = x + \frac{r}{2}t, \tag{3.4}$$

and impose

$$f_{24}(M) = b, \qquad f_{24}(S) = b - \frac{\kappa r^3}{6}, \qquad \nabla f_{24}(M) = \nabla f_{24}(S) = 0, \tag{3.5}$$

with $\kappa > 0$. The change from the saddle value $h = f_{24}(S)$ to κ has the exact Jacobian

$$|dh| = \frac{r^3}{6}\, d\kappa. \tag{3.6}$$

Use the corrected divided-difference pin frame

$$V^+ = \frac{f(S)+f(M)}{2}, \qquad V^-_{\mathrm{corr}} = \frac{f(S)-f(M)-\frac{r}{2}(\partial_t f(S)+\partial_t f(M))}{r^3}, \tag{3.7}$$

$$G^+_e = \frac{\partial_e f(S)+\partial_e f(M)}{2}, \qquad G^-_e = \frac{\partial_e f(S)-\partial_e f(M)}{r}, \tag{3.8}$$

for $e \in \{t, n_1, n_2\}$. The exact determinant of the raw-to-corrected pin map is r^{-6}. Hence the raw pin density contributes r^{-6}.

At a generic fold, one eigenvalue of each endpoint Hessian is soft. Conditionally on the pins,

$$\det H_M = rD_M, \qquad \det H_S = rD_S, \tag{3.9}$$

with conditional Gaussian moments bounded by a polynomial in $|b|+\kappa$, uniformly in t and in r below any fixed $\varrho < 12$, and hence uniformly bounded on compact mark sets (the proof of Supplement Proposition S6.4); consequently the two endpoint determinant weights contribute r^2.

Combining (3.3), (3.6), the corrected-pin density and (3.9) gives

$$\frac{r^3}{6}\, r^{-6}\, r^2\, r^2\, dr = \frac{1}{6}\, r\, dr. \tag{3.10}$$

No additional ordering, antipodal, branch or merge multiplicity occurs.

3.3. **Lifetime pushforward.** The lifetime in (3.5) is

$$\ell = \frac{\kappa r^3}{6}. \tag{3.11}$$

For fixed $\kappa > 0$,

$$r\, dr = \frac{6^{2/3}}{3}\, \kappa^{-2/3}\, \ell^{-1/3}\, d\ell. \tag{3.12}$$

Equations (3.10)–(3.12) already determine the exponent $-1/3$. It remains to prove that elder selection has a nonzero limiting mass and to evaluate the coefficient. Fix $\varrho \in (0, 12)$ and split $\nu_{3,24} = \nu^{\mathrm{near}}_\varrho + \nu^{\mathrm{far}}_\varrho$ according to whether the elder pair has separation $r < \varrho$ or $r \geq \varrho$.

Proposition 3.3 (exact near pushforward; Supplement Proposition S6.1)**.** *Put $G^{\mathrm{sel}}_r = r^4 K^{\mathrm{sel}}_r$ and $G^{\mathrm{all}}_r = r^4 K^{\mathrm{all}}_r$. For every nonnegative test function ψ,*

$$\int \psi(\ell)\, \nu^{\mathrm{near}}_\varrho(\ell)\, d\ell = \int_{S^2} \int_{\mathbb{R}} \int_0^\infty \int_0^\varrho \psi(\kappa r^3/6)\, K^{\mathrm{sel}}_r\, \frac{r^3}{6}\, r^2\, dr\, d\kappa\, db\, d\sigma_2(t), \tag{3.12a}$$

and therefore, with $r_\ell(\kappa) = (6\ell/\kappa)^{1/3}$,

$$\nu^{\mathrm{near}}_\varrho(\ell) = \frac{6^{2/3}}{18}\, \ell^{-1/3} \int_{\{\kappa > 6\ell/\varrho^3\}} \kappa^{-2/3}\, G^{\mathrm{sel}}_{r_\ell(\kappa)}\, d\kappa\, db\, d\sigma_2(t). \tag{3.12b}$$

3.4. **Elder selection.** Let $p_r(t, b, \kappa)$ be the Palm probability (Lemma 3.2) that the typed maximum–index-two-saddle contact pair in (3.5) is the actual elder persistence pair.

Lemma 3.4 (trichotomy; Supplement Lemma S5.7)**.** *On the Morse event, for an ordered typed pair (M, S) at distance r with midpoint x,*

$$A_r \cap E_r^c \subset B_r \cup L_{\mathrm{same},r},$$

where A_r is gradient adjacency, $E_r = \{e = 1\}$, $L_{\mathrm{same},r}$ is same-maximum self-attachment (both ascending half-branches of S end at M), and B_r is the existence of a third critical point with value in the pair window $[h, b]$ in one of three spatial regions measured from x: the collar $\{|y-x| \leq C_1 r\}$, the singular-near region $\{C_1 r < |y-x| \leq \delta\}$, or at fixed distance $\{|y-x| > \delta\}$. If the two upper sectors of S lie in one component of $\{f > h\}$ and not both half-branches end at M, that component contains a critical point other than M and S with value in (h, b) (Supplement Lemma S5.7, Case 0,

with the merge-level Lemma S5.6), so that case lies in B_r. When M is the older birth maximum but $e = 0$, the witness is the younger component's own birth maximum M_2, with $h < f(M_2) < f(M)$.

Corollary 3.5 (the selection loss has exact order r^3)**.** *Uniformly for $t \in S^2$ and compact $b, \kappa > 0$,*

$$1 - p_r(t, b, \kappa) \le Cr^3, \tag{3.13}$$

and, for constants $c, r_* > 0$ depending on the compact mark set,

$$1 - p_r(t, b, \kappa) \ge c\, r^3, \qquad 0 < r \le r_*. \tag{3.13a}$$

Corollary 3.5 is proved after Theorem 3.6; its upper bound (3.13) is Supplement Theorem S12.3, which states Theorem A of [R2] and proves it in Supplement Part III. The rest of this subsection describes a second, three-dimensional route to (3.13) through explicit witnesses, written out in Supplement §S5 (Theorem S5.36). A provider-distinct read of the eighth draft of that section found a gap in it: the collar and singular-near counts (Supplement Lemmas S5.19 and S5.23) passed from contact-limit densities to the densities at a fixed positive scale without a proof that is uniform near the pair axis. Supplement draft 9 repairs it in §S5.8 (Lemmas S5.39–S5.41), by a radial blow-up of the parameters at the axis with explicit limit functionals, and proves the two counts again from it (§7 records the reads of this repair). The proof of the corollary does not use this route.

The three witness families are controlled as follows.

First, the collar and singular-near third-critical-point charts retain their exact vanishing factors under the side-24 periodized jet (Supplement Lemmas S5.18 and S5.21). In collar variables $y = x + r(Xt + \rho v)$, with v a unit vector orthogonal to t, the scaled conditional gradient covariance factors as

$$\det C_{\mathrm{col}} = \rho^6 (X^2 + \rho^2) A_{\mathrm{col}}, \qquad A_{\mathrm{col}} > 0. \tag{3.14}$$

In the singular-near chart $y = x + s(ct + \rho v)$, with $c^2 + \rho^2 = 1$ and $s = |y - x|$, the corrected value–gradient covariance factors as

$$\det C_{\mathrm{sing}} = \rho^{10} (3c^2 + \rho^2) A_{\mathrm{sing}}, \qquad A_{\mathrm{sing}} > 0. \tag{3.15}$$

Together with the anisotropic triple-determinant envelope

$$[r(r + s^2)]^2 (r^2 + s^3), \tag{3.16}$$

which bounds $|\det H_M \det H_S \det H_y|$ up to $C\rho^{-6}$ times the cube of a derivative bound (Supplement Lemma S5.22), these give Palm expectations $O(r^3)$ for collar and singular-near witnesses (Supplement Lemmas S5.19 and S5.23), with the finite-scale densities of Supplement §S5.8. The envelope is proved from Taylor's formula, the pins and the value window, uniformly in the angle. It does not use the per-Hessian formulas of the exact triple-contact elimination (Supplement Lemma S5.15; Remark 3.8).

Second, complete positive Fourier support of (1.1) implies strict covariance positivity for every linearly independent finite family of derivative-evaluation distributions. Compactness then gives uniform eigenvalue floors for the corrected pair pins joined to a fixed-distance remote value-gradient block. The endpoint determinant factor r^2, the remote value window of width $\kappa r^3/6$ and the Palm normalizer $Z_r \asymp r^2$ give, uniformly for $t \in S^2$ and compact $b, \kappa > 0$,

$$\mathbb{E}_r^{MS} N_{\mathrm{far}} \le Cr^3. \tag{3.17}$$

The count is conditional on the pins, so the argument needs the pin density to be bounded below; it gives no Gaussian factor in (b, κ).

Third, the local unstable-manifold comparison is performed in the correctly scaled transverse matrix

$$T_r = -\frac{Q_\perp}{\kappa r}, \tag{3.18}$$

where $Q_\perp$ is the physical transverse Hessian at the contact saddle (Supplement (S5.95)). The inward branch is captured by the maximum, and the outward branch crosses a forward section at a level strictly above $f(M)$; this is Theorem 3.6. Shallow-eigenvalue, finite-jet and fifth-derivative failures have Palm probability $O(r^3)$ or $o(r^3)$.

Theorem 3.6 (energy-adapted capture and escape; Supplement Theorem S5.31)**.** *On a fixed compact convex chart, with $X \in \mathbb{R}$ along the pair and $Y \in \mathbb{R}^2$ across it, consider the normalized potential*

$$P(X,Y) = p_0(X) + \tfrac{1}{2}(X^2 - \tfrac{1}{4})a \cdot Y - \tfrac{1}{2}Y^\top TY + \tfrac{1}{2}X\,Y^\top WY + \tfrac{1}{6}C[Y,Y,Y] + r\,Q(X,Y), \quad (3.18a)$$

with $p_0(X) = X^3/3 - X/4 - 1/12$, $T \succ 0$ and W symmetric, C a symmetric cubic tensor, and Q with $Q = 0$ and $\nabla Q = 0$ at $M = (-\frac{1}{2}, 0)$ and $S = (\frac{1}{2}, 0)$; for the conditioned side-24 field, $T = T_r$ of (3.18). Put $R = 1 + |a| + \|W\| + \|C\| + \|Q\|_{C^2}$ and $\tau = \lambda_{\min}(T)$, and fix $K = 4096$, $\varepsilon_0 = 1/(1024K)$. If $rR^5 \le \varepsilon_0$, $\tau \ge 2KR^5 + 2R$ and $r \le r_{\mathrm{mat}}$, the explicit radius of Supplement (S5.94), then:

- *(i) the M-ward unstable half-branch of S converges to M through the tapered tube $|Y| \le (2R/\tau)(\frac{1}{4} - X^2)$;*
- *(ii) the outward branch stays in an energy cone and strip and reaches $X = 5/4$ at a level greater than $49/192 - 1/768 = 65/256 > 1/4$, strictly above the level of M, so it cannot terminate at M;*
- *(iii) both conclusions hold for the exact conditioned field on the fifth-derivative good event.*

No bound on $\lambda_{\max}(T)$ is assumed or used. The four numerical margins of the proof are verified in exact rational arithmetic.

Supplement §S5.5 proves Theorem 3.6. Its proof contains five local repairs of detail to the version of 1 August 2026, none of which changes the statement: the bound on $r\,Q_X$, the sign of the tangent eigenvector, containment near S, the position of the capture section, and the rQ part of the energy remainder.

Proof of Corollary 3.5. The bound (3.13) is Supplement Theorem S12.3 at $d = 3$ and $L = 24$; it is Theorem A of [R2], with its proof in Supplement Part III. That theorem is proved for every fixed $d \ge 2$ and $L > 0$, uniformly over compact birth and gap windows and over all orthonormal frames, with existential constants. Its selection probability is the probability that the global superlevel elder partner of M is S, which is p_r. The witness route above outlines a second, three-dimensional argument (Supplement Theorem S5.36), with the repair noted at the start of this subsection: by Lemma 3.4, selection failure produces a collar witness, a singular-near witness, a fixed-distance witness or a same-maximum self-attachment, and by the three estimates above and Theorem 3.6 each of these events has Palm probability $O(r^3)$. For (3.13a): on an event of Palm probability at least cr^3 — the planar polygonal-path obstruction, embedded in the soft eigenplane of the pair with the remaining transverse directions hard — the component born at M is already joined, within $\{f > f(S)\}$, to a component with an older birth maximum, so the elder rule rejects the pair [R6]. □

By (3.12), the elder-selected lifetime density on any compact mark window therefore differs from the all-typed one by exactly $\Theta(\ell^{2/3})$. The leading constant $c_{3,24}$ is unaffected. The constants in (3.13)–(3.13a) are existential.

Remark 3.7 (the rejection mechanism)*.* The mechanism of rejection is local and explicit. At scale r the pair sees, in the limit, a pinned planar cubic; in that limiting model the pair is rejected exactly when the cubic carries an extra critical point at a height between $f(S)$ and $f(M)$, in which case M dies at the highest such saddle. For the actual field this decision transfers from the model only up to exceptional sets and under the conditions recorded in §§5.3 and 5.5. The coefficient of the

$\Theta(\ell^{2/3})$ selection loss is then a Gaussian integral of the corresponding local cluster mass [R10]; see also §5.3.

Remark 3.8 (the triple-contact elimination). An independent exact elimination [R1] confirms the first line of the display (B.1) of 1 August 2026 and corrects two transcription slips. In the variables of Supplement Lemma S5.15 (c and ρ the cosine and sine of the angle between t and $y-x$, $\eta = r/|y-x|$, and q_M, q_S the in-plane transverse curvatures at the endpoints), the κ-linear bracket of $-\det H_S$ carries a minus sign,

$$-\det H_S = \frac{\kappa^2(4c^2-\eta^2)^2 - 4\kappa\rho^2\big((4c^2-\eta^2)q_M + (12c^2+\eta^2)q_S\big) + 4\rho^4(q_M-q_S)^2}{64c^2\rho^2},$$

and the κ-free term of the numerator N_y of $\det H_y = N_y/(64c^2\eta\rho^2)$ is $-16c(q_M^2 - q_S^2)\rho^4 + 4\eta(q_M - q_S)^2\rho^4$. Both κ^2 coefficients equal $(4c^2-\eta^2)^2/(64c^2\rho^2)$. Supplement Lemma S5.15 prints the corrected formulas. The envelope (3.16) does not use them.

3.5. **All-mark domination, the near/far decomposition and the limit.** The corrected pair target is uniformly injective in (b,κ). Gaussian regression and bounded conditional polynomial moments give the common majorant

$$C\big[1 + (|b|+\kappa)^N\big]e^{-c(b^2+\kappa^2)}\,\kappa^{-2/3}, \tag{3.19}$$

valid for all $0 < r < \varrho$ and integrable on $S^2 \times \mathbb{R} \times (0,\infty)$. Since $0 \le p_r \le 1$, (3.13) and dominated convergence show that the selected and all-typed contact integrals have the same limit.

Theorem 3.9 (limit interchange; Supplement Theorem S6.3)**.** *Assume (H1) $G_r^{\mathrm{sel}} \to G_0^{\mathrm{sel}}$ pointwise as $r \downarrow 0$; (H2) one integrable majorant for $\kappa^{-2/3}G_r^{\mathrm{sel}}$, $0 < r < \varrho$, including $\kappa \downarrow 0$; (H3) $G_0^{\mathrm{sel}} = G_0^{\mathrm{all}}$, the pointwise limit (3.19a) of G_r^{all}. Then*

$$\nu_\varrho^{\mathrm{near}}(\ell) = c_{3,24}\,\ell^{-1/3}(1+o(1)), \qquad c_{3,24} = \frac{6^{2/3}}{18}\int \kappa^{-2/3}\,G_0^{\mathrm{all}}\,d\kappa\,db\,d\sigma_2(t).$$

All three hypotheses are verified in Supplement Proposition S6.4: the majorant (3.19) supplies (H2), and the contact limit (3.19a) together with (3.13), which gives $p_r \to 1$, supplies (H1) and (H3). The same proposition gives $0 < c_{3,24} < \infty$.

Proposition 3.10 (far elder pairs; Supplement Proposition S6.2)**.** *For fixed $\varrho \in (0,12)$ there is $\ell_0 > 0$ such that $\sup_{0<\ell\le\ell_0}\nu_\varrho^{\mathrm{far}}(\ell) < \infty$. In particular $\nu_\varrho^{\mathrm{far}}(\ell) = o(\ell^{-1/3})$.*

Proof. This is Supplement Proposition S6.2. Its proof bounds the elder mark by one and applies the marked two-point Kac–Rice formula (Supplement Proposition S12.22) on the compact set of pairs at distance at least ϱ, where the value–gradient covariance is nonsingular; it cites no result of [R2] that the Supplement does not itself prove. For $0 < \ell \le 1$ and pairs at distance at least the small radius r_0 of [R2], the same bound is its (14.1) (Supplement Lemma S12.29). □

Theorem 3.9 and Proposition 3.10 give the leading term of (2.1). Independently, the far elder contribution tends to zero as $\ell \downarrow 0$ [R9]. Two rates are on record [R9]: $O(\ell^{2/3})$, because a bar of lifetime ℓ is born at a maximum of weakest curvature $O(\ell^{1/3})$ (reviewed and merged; proved in Supplement Proposition S8.13); and $O(\ell^N)$ for every N, because on the superlevel component of a living bar the gradient is $O((K(b-f))^{1/2})$, so that a far death forces N separated near-critical sites (one provider's nonauthor review). A reviewed quantitative extension allows the separation to shrink like a small power of ℓ [R9]. The bound $O(\ell)$ asserted for the far density in the amendment of 1 August 2026 (its Proposition A.3.2) is withdrawn: its proof bounded the far *candidate* density, which does not vanish (Remark 2.5).

3.6. **Exact coefficient reduction.** Conditionally on the pins, as $r \downarrow 0$, we have $r^{-1}\det H_M \to -\kappa \det Q_t(q_t)$ and $r^{-1}\det H_S \to \kappa \det Q_t(q_t)$, and the type indicators converge to $\mathbf{1}_{\{Q_t(q_t)\prec 0\}}$ (Supplement Proposition S2.8). The corrected pins (3.7)–(3.8) have the contact limit $Y_t = (f, u_t, g_t, h_t)$, with target $(b, -\kappa/6, 0, 0)$. Hence G_r^{all} has the pointwise limit

$$G_0^{\mathrm{all}}(t, b, \kappa) = p_{Y_t}(b, -\kappa/6, 0, 0)\, \kappa^2\, \mathbb{E}\big[F(q_t) \mid Y_t = (b, -\kappa/6, 0, 0)\big] \tag{3.19a}$$

(Supplement (S3.27)), where $F(q) = (\det Q_t(q))^2 \mathbf{1}_{\{Q_t(q)\prec 0\}}$. Even and odd covariance derivatives are independent because K_{24} is even, so (3.19a) is the product of $p_{(f,h_t)}(b,0)\,\mathbb{E}[F(q_t) \mid f = b,\ h_t = 0]$ and $p_{(u_t,g_t)}(-\kappa/6, 0)\,\kappa^2$. Integrating over the birth height removes the field-value coordinate:

$$\int_{\mathbb{R}} p_{(f,h_t)}(b,0)\, \mathbb{E}\big[F(q_t) \mid f = b,\ h_t = 0\big]\, db = p_{h_t}(0)\, \mathbb{E}\big[F(q_t) \mid h_t = 0\big]. \tag{3.20}$$

Conditionally on $g_t = 0$, the variable u_t is centered Gaussian with variance σ_t^2. The remaining normalized-gap integral, with the weight $\kappa^{-2/3}$ of Theorem 3.9, is

$$\int_0^\infty p_{(u_t,g_t)}(-\kappa/6, 0)\, \kappa^{4/3}\, d\kappa = p_{g_t}(0)\, \frac{\Gamma(7/6)\, 72^{7/6}\, (\sigma_t^2)^{2/3}}{2\sqrt{2\pi}}. \tag{3.21}$$

Inserting (3.19a), (3.20) and (3.21) in the formula for $c_{3,24}$ of Theorem 3.9 gives exactly (2.9).

Equivalently, if $A_t A_t^\top = \Gamma_t$ and $\mathcal{Q}_t(v) = Q_t(A_t v)$, then

$$D_t = \frac{15}{4\pi} \int_{S^2} [\det \mathcal{Q}_t(v)]^2\, \mathbf{1}_{\{\operatorname{tr} \mathcal{Q}_t(v) < 0,\ \det \mathcal{Q}_t(v) > 0\}}\, dv. \tag{3.22}$$

Thus $c_{3,24}$ is a deterministic nested angular integral involving only the order-six covariance jet at the origin.

3.7. **The value on $\mathbb{R}^3$.** At the jet of the (non-periodized) Bargmann–Fock covariance $e^{-|z|^2/2}$,

$$\Sigma_g = I_3, \qquad \Sigma_h = \operatorname{diag}(3, 1, 1), \qquad \sigma_t^2 = \frac{1}{24}, \tag{3.23}$$

and

$$Q \mid h_t = 0 \ \overset{d}{=}\ G_2 + \sqrt{\tfrac{2}{3}}\, Z\, I_2, \qquad D_t = \frac{29}{6} - \sqrt{6}, \tag{3.24}$$

where G_2 is a 2×2 GOE matrix (diagonal entries $N(0,2)$, off-diagonal entry $N(0,1)$, all independent) and $Z \sim N(0,1)$ is independent of G_2. The value of D_t is proved in closed form in Supplement Theorem S3.16: in trace and difference coordinates $T \sim N(0,4)$ is independent of $\Delta \geq 0$, $\det Q = (W^2 - \Delta^2)/4$, and the cone condition is $W < -\Delta$ for the single Gaussian $W = T + 2\sqrt{2/3}\, Z \sim N(0, 20/3)$. Substitution in (2.9) gives (2.11).

3.8. **Side-24 correction.** Let J_0 be the even covariance jet of $e^{-|z|^2/2}$ through order six, J_{24} the corresponding jet of (1.1), and

$$F(J) = \frac{\Phi(J)}{\Phi(J_0)}, \tag{3.25}$$

where Φ is the coefficient functional in (2.9). The bound $|\varepsilon_{24}| < 10^{-216}$ in (2.10) is proved in Supplement Theorem S4.4, which follows [R4]; see also the end of this subsection. Displays (3.26)–(3.33) outline a shorter route to the weaker bound 10^{-180}. Its bounds (3.29) are Supplement Lemma S4.29, while the derivative bounds (3.27) are quoted from the technical appendix of 30 July 2026 (Modules D and H) and are not reproved in the Supplement. On the explicit jet ball

$$\|J - J_0\|_{\max} \leq 10^{-110}, \tag{3.26}$$

the covariance and Schur-complement matrices have uniform positive lower and finite upper bounds. Fixed-cone Gaussian differentiation gives

$$\sup \|DF\| < 10^{16}, \qquad \sup \|D^2 F\| < 10^{42}. \tag{3.27}$$

Put $q = e^{-288}$. The normalized periodized jet has the decomposition

$$J_{24} - J_0 = J^{(1)}_{\mathrm{near}} + R_{\mathrm{jet}}, \tag{3.28}$$

where the linear term $J^{(1)}_{\mathrm{near}}$ is the jet of $2e^{-288}e^{-|z|^2/2}\sum_{i=1}^{3}(\cosh(24z_i)-1)$: the six images $\pm 24e_i$ together with their share $6e^{-288}$ of the normalizer Z_{24} (Supplement Definition S4.22). Exact rational Hermite bounds give (Supplement Lemma S4.29)

$$\|J_{24} - J_0\|_{\max} < 10^{-113}, \qquad \|R_{\mathrm{jet}}\|_{\max} < 10^{-230}. \tag{3.29}$$

Rotational projection of the six-image orbit yields

$$DF(J_0)\big[J^{(1)}_{\mathrm{near}}\big] = P_3(24)\, e^{-288}, \tag{3.30}$$

where

$$P_3(L) = -\frac{L^2(10L^4 - 147L^2 + 315)}{105}, \qquad P_3(24) = -\frac{620813376}{35}. \tag{3.31}$$

The first variation (3.30) is derived from first principles in Supplement Proposition S4.24 and Corollary S4.25. In the jet coordinates a, m_4, χ, Ω of the technical appendix (Module H; Supplement Lemma S4.20 and Corollary S4.25), the variations per unit $e^{-L^2/2}$, $\delta a = -2L^2$, $\delta m_4 = 6L^4/5 - 12L^2$ and $\delta\chi = -6L^6/7 + 18L^4 - 90L^2$ and the display $\delta \log c = \delta\Omega/9 + \delta m_4/6 - \frac{13}{6}\delta a = P_3(L)$ follow from kernel differentiation of the periodized kernel and Haar projection, with the pure partials $\partial_a = \partial_{m_4} = -1/2$, $\partial_\chi = 1/9$ taken through the full nonlinear functional (including the homogeneity term of $D(\Gamma)$), and the scale check $d\log c/d\log\lambda = 3/2$ for the kernel $e^{-\lambda|z|^2/2}$ (Supplement Remark S4.26); the regrouped display is an algebraic identity via $\delta\Omega = \chi\,\delta a + a\,\delta\chi - 2m_4\,\delta m_4$. Both $P_3(24) = -620813376/35$ and the first variation are also derived independently in [R4]. Taylor's theorem, (3.27) and (3.29) give, with R_{quad} the second-order Taylor remainder of F at J_0 along $J_{24} - J_0$,

$$|DF(J_0)[R_{\mathrm{jet}}]| < 10^{-210}, \qquad |R_{\mathrm{quad}}| < 5\cdot 10^{-185}, \tag{3.32}$$

and therefore, as a bound weaker than (2.10),

$$F(J_{24}) - 1 = P_3(24)\, e^{-288} + \varepsilon_{24}, \qquad |\varepsilon_{24}| < 10^{-180}. \tag{3.33}$$

The second-order remainder admits two sharper and independent bounds [R4]. (i) A second-order Taylor expansion of the coefficient functional along the periodization perturbation, using a Gaussian-section representation of the coefficient as an angular integral and an exact Haar first order, gives $|\varepsilon_{24}| < 10^{-218}$, with a remainder ledger below $6\cdot 10^{-235}$. (ii) A derivation with exact χ^2_ν radial moments gives $|\varepsilon_{24}| < 10^{-216}$ for $d = 2, 3$ (Supplement Theorem S4.4), and identifies the general polynomial

$$P_d(L) = -\frac{10L^6}{3(d+2)(d+4)} + \frac{(d+4)L^4}{d+2} - dL^2,$$

which reduces to P_3 at $d = 3$. Either bound gives (2.10); we state the weaker one. This completes the proof of the leading term of Theorem 2.1 and of (2.10).

Remark 3.11 (torus and reference contact coefficients). The near-contact quantities themselves move very little under periodization. On the torus of side $L = 24$, for birth heights in $[0, 1]$ and gaps in $[1/2, 2]$, the $d = 3$ near-contact numerators and normalizer differ from their reference-kernel values by less than 10^{-26}, and the compact-band contact integral differs by less than 10^{-19} in absolute value [R20].

3.9. **The rate.**

Proposition 3.12 (bounded remainder)**.** *There are $C, \ell_* > 0$ such that $|\nu_{3,24}(\ell) - c_{3,24}\ell^{-1/3}| \le C$ for $0 < \ell \le \ell_*$.*

Proof. This is Theorem R of [R3] at $d = 3$ and $L = 24$, proved in Supplement Theorem S7.9; Supplement Remark S7.22 identifies its density and its constant with those of Theorem 2.1. Theorem R holds for every fixed $d \ge 2$ and $L > 0$, for all marks and separations, with existential constants. □

Proposition 3.12 is the rate in (2.1). In Theorem R the near-contact error, the near pairing loss, and the far pairs each contribute $O(1)$ to that bound, and Theorem R does not prove that the remainder tends to zero. The far elder density tends to zero at the rates above; the rejected density tends to a positive constant. Theorem 2.3 identifies the next terms of the elder density. □

4. Exact scope

Theorem 2.1 is restricted to:

- ambient dimension 3;
- torus side 24;
- the normalized periodized Bargmann–Fock covariance (1.1);
- finite nonessential superlevel H_0 bars;
- the first-moment lifetime density as $\ell \downarrow 0$;
- the ordered maximum–index-two-saddle, directed-S^2 and Palm conventions used above.

It does not imply:

- a theorem for other side lengths;
- a theorem for arbitrary isotropic or stationary Gaussian fields;
- a result for H_k, $k > 0$;
- an interchange of the limits $L \to \infty$ and $\ell \downarrow 0$;
- an infinite-volume persistence process;
- an explicit range of ℓ on which (2.1) or (2.12) holds to a stated accuracy: every remainder constant in this paper is existential;
- a numerical value of ε_{24}, or a directed-rounding certificate for the nonlinear integral (2.9) beyond the relative bound $|c_{3,24}/c_{3,\infty} - 1| < 10^{-106}$ [R5]. Such a certificate is not used in the analytic estimate (2.10).

The records behind Theorem 2.3 and Remarks 2.4–2.5 hold in the dimensions and sides stated in each record, at their stated scopes, and the numerical enclosures are specific to the sides stated. This paper states them for $d = 3$ and $L = 24$, except where a statement names other sides or dimensions: the transfer of c_2 in Theorem 2.3(iii) holds for every $L \ge 24$, and §3.8 records the polynomial $P_d(L)$ for every d.

5. Discussion

Statements in this section are labelled *proved* (a merged record with nonauthor review), *read* (read in bounded nonauthor slices from the same account), *author-side*, *numerical* (computation or simulation without certification) or *expectation* (not proved); conditional scopes are stated in place. None is used in §3.

5.1. **Why $-1/3$, and what depends on the law.** In the separation variable $r = |M - S|$ and the height gap $\ell = f(M) - f(S)$, the ordered maximum/saddle pair measure is the polar volume r^{d-1} times the two-site pin density, of order $r^{-(d+3)}$, times the determinant scale r^2; that is, $r^{-2}\,dr\,d\ell$ in every dimension. A pinned pair has gap $\ell = kr^3$, so $d\ell = r^3\,dk$ at fixed r and the measure is $r\,dr\,dk$;

this is (3.10) with $k = \kappa/6$. Pushing forward to the lifetime at fixed k, $r\,dr = \frac{1}{3}k^{-2/3}\ell^{-1/3}\,d\ell$ (cf. (3.12)), which gives $\ell^{-1/3}$ with no dependence on d; the corresponding theorem of [R2] (Theorem C) holds for every $d \geq 2$, and the present $d = 3$, side-24 statement is one instance. The exponent is the Whitney-fold exponent: in a one-parameter family a maximum/saddle pair annihilates through $f_t(x) = x^3/3 - (t_0 - t)x$, with separation proportional to $(t_0 - t)^{1/2}$ and gap proportional to $(t_0 - t)^{3/2}$, so the time spent with gap in $d\ell$ is proportional to $\ell^{-1/3}\,d\ell$. The Gaussian law enters only the constant: by [R2], (15.2), $c_{d,L}$ is an angular average of $p_G(0)\,p_{V_u}(0)\,\tau_u^{4/3}D_u$, that is, of densities and of a cone moment of the 3-jet at the collision point, and it scales as $a^{-2/3}$ under $f \mapsto af$. The proof separates accordingly: the elder rule near a collision (the cap theorem), the barrier $\lambda_{\min}(-D^2 f(M)) \leq (\ell K^2/c_L)^{1/3}$ for a bar of lifetime ℓ, and the pinned-pair identities are deterministic statements about C^3 functions; the Gaussian structure supplies the regression couplings and the uniform moment and density bounds. We therefore expect, but do not prove, that the exponent is shared by stationary fields with C^7 paths, finite moments of the derivative suprema and a continuous positive density of the 3-jet at a point, with c given by the analogue of (15.2) of [R2] (*expectation*).

The near-diagonal law has a second scale. The leading term comes from separations $r \asymp \ell^{1/3}$; at $r \asymp \ell^{1/4}$ the cubic coefficient is comparable to the separation ($\ell \asymp r^4$, the cusp organizing the fold family), and there the first correction arises: on that scale the pinned landscape is a universal quartic ridge on which a maximum/saddle pair of the right types is an elder pair in only one third of its type window, so the local contact law overcounts and the correction $c_1\ell^{1/4}$ of Theorem 2.3 is negative.

5.2. **The corrections.** At separations $r \asymp \ell^{1/4}$ the rescaled pinned landscape converges to a universal random quartic whose superlevel topology reduces exactly to the one-dimensional ridge

$$g = \vartheta\big[2(X + \tfrac{1}{2})^2(X - 1) + 3\varphi(X^2 - \tfrac{1}{4})^2\big];$$

where $\vartheta = \ell/r^4$ is the cusp variable (the records [R11] and [R15] write it κ; here κ keeps its meaning from (3.5), so $\vartheta = \kappa/(6r)$); the pair is typed if and only if $|\varphi| < 1$, and the saddle is the maximum's elder partner if and only if $|\varphi| < 1/3$ (at $\varphi = \pm 1/3$ two competing saddles, respectively maxima, have equal heights). Integrating the resulting loss over the scale gives (2.13) (*proved*, [R11]). Through the order $\ell^{1/3}$, the elder rule enters the expansion only through this cusp scale: there the decision is stable with an explicit margin proportional to $||\varphi| - 1/3|$, while at the fold scale and at larger separations the elder rule costs a power of ℓ beyond $\ell^{1/3}$. So the third coefficient of the elder law is the same fold-scale constant c_2 as for the candidate law (*proved*, [R11]). Fold-scale rejection does enter at the next order, through the term $R_{2/3}\ell^{2/3}$ of Remark 2.4.

At large ϑ the rejected pairs are soft: exactly one transverse curvature is small, of order γ_1^2/ϑ, and the rejected cusp kernel decays like F_0/ϑ in every dimension; for the Gaussian kernel in $d = 3$, $\iint F_0 = 125\sqrt{30}/(192\pi^3) \approx 0.1150$ (*proved*, [R15]). The fold-scale rejection rate is the selection-loss coefficient of Remark 3.7; for the Gaussian kernel it equals $F_0 H(k)$ with $H(k) = 1 + \frac{12}{25}k^2 + h_{7/2}k^{7/2} + O(k^4)$ and an explicit $h_{7/2} < 0$ (*proved* at the conditional scope of [R15], Theorem B). These soft rejected pairs produce the $\ell^{2/3}$ term $R_{2/3}$ of the rejected density in Remark 2.4; in a full-field simulation in $d = 3$, rejected adjacent pairs with $r/\ell^{1/4} < 0.8$ number 11, against 10.7 predicted and 1,329 from the cusp kernel alone (*numerical*, [R16, R15]).

Size of the corrections. By (2.14), the total relative correction of the three-term law is about -1.5% at $\ell = 10^{-4}$, -5% at 10^{-3} and -17% at 10^{-2}; the two corrections partially cancel. A direct simulation of the elder-rule persistence of the periodized Gaussian field (sides 64 in $d = 2$ and 16 in $d = 3$, not 24; at such sides the periodization changes the coefficients by terms of order at most $L^8 e^{-L^2/2}$, as in (2.10) and Supplement Corollary S11.10), with exact critical values and exact

one-dimensional connectivity by ascending lines, agrees with the three-term law in aggregate over $10^{-4} \le \ell \le 10^{-2}$ (data over law 1.006 ± 0.004 in $d = 2$ and 1.008 ± 0.010 in $d = 3$) and rejects the leading and two-term laws; it also shows the cusp-scale elder window $|\varphi| < 1/3$ directly. Bin by bin the three-term law does not fit exactly: over 14 bins on $[10^{-4}, 3 \cdot 10^{-2}]$, χ^2 is 84 in $d = 3$ and 44 in $d = 2$. Adding the $\ell^{2/3}$ term of Remark 2.4 reduces these to 23 and 7, leaving a small negative next-order term (*numerical*, [R16]). The simulation illustrates the law; it does not prove it, and its residuals are not evidence for an $\ell^{1/2}$ term.

5.3. **Selection loss in dimension three.** In $d \ge 3$ the hard transverse directions do not change the local decision: maximizing over the hard fibre reduces the decision window to the planar model up to an error of order r in C^2, so the planar elder/rejected certificate transfers to the actual field with margins of the same order (*author-side*: addenda A3 and A3.1 with their erratum, read in bounded nonauthor slices, [R18]). On compact marks the exhaustion has been carried out in $d = 3$: the failure-measure rate of the records is r^β for every $\beta < 1$, with

$$1 - p_r = r^3 \alpha^{(3)} + O(r^4 \log(1/r)^7),$$

where $\alpha^{(3)}$ is the local cluster mass of Remark 3.7, identified with the fold-scale coefficient $F/(kA_0)$ of [R10]. As a corollary, on compact windows of births and gaps the rejected density is $C^{(3)}_{\mathrm{fail}} \ell^{2/3} + O(\ell \log(1/\ell)^7)$, with $C^{(3)}_{\mathrm{fail}} = \int A_0 \, \alpha^{(3)}/(3k^{5/3}) \, db \, dk \, d\sigma(u)$ over the window and A_0 the pinned-pair weight of [R10] (*read*, and conditional on the author-side chain A3.3–A3.8; independence credit 0; [R18]). All constants are existential, and no rate of this type is claimed for the unrestricted population.

5.4. **Dimension one.** In dimension one every ingredient of the short-bar expansion can be made explicit. For a smooth stationary Gaussian process on a circle, and on the line under a mild spectral hypothesis, the persistence lifetime density satisfies

$$\nu(\ell) = C_0 \ell^{-1/3} + C_1 \ell^{1/4} + 2B_2 \ell^{1/3} + O(\ell^{3/4}),$$

with $C_1 < 0$ and B_2 explicit in the spectral moments; $B_2 > 0$ when $\lambda_2 \lambda_6 \le 25\lambda_4^2$, for instance for the Gaussian kernel, but its sign is not fixed in general (*proved* on the circle, Theorems 1D and 1D$^+$; on the line *author-side*, read by four nonauthor slices and merged; [R17]). The coefficient C_1 is the one-dimensional value of the cusp formula (2.13), the elder window of the cusp ridge is the same $|\varphi| < 1/3$, and the remainder is of the order of the first correction to the cusp term. For the covariance $e^{-x^2/2}$ the crest-to-trough amplitude density per crest, which counts every adjacent crest–trough pair and is therefore the one-dimensional candidate law, is $0.19971814\, h^{-1/3} - 0.27165891\, h^{1/4} + 0.41725471\, h^{1/3} + O(h^{3/4})$. A simulation over $8.4 \cdot 10^7$ correlation lengths reproduces the three-term law to 0.3% and the coefficient of the rejected pairs to 1% (*numerical*). For the Gaussian kernel the second- and third-order terms of this amplitude density nearly cancel for $10^{-4} \le h \le 10^{-2}$ (relative coefficients -1.36 and $+2.09$; net relative correction below 0.5% there). Those of the lifetime density, with relative coefficients $C_1/C_0 = -2.07$ and $2B_2/C_0 = +2.09$, cancel only in part: the net relative corrections are -0.5%, -1.6% and -4.4% at $\ell = 10^{-4}$, 10^{-3} and 10^{-2}. This is a caution against reading corrections off histograms at moderate ℓ.

5.5. **The plane.** In dimension two the local pairing rule is established for the actual Gaussian field with a rate. On the bounded soft layer, the H_0 pairing of the designated pin pair agrees with the model's elder decision except on a set of weighted probability $O(r^{7/2})$, against a layer mass of order r^3 (*proved*, [R19]). On compact gap intervals the selection loss satisfies $1 - p_r = r^3(\alpha_1 + \alpha_2) + O(r^4 \log(1/r)^4)$, so that on compact windows of births and gaps the rejected density

is $C_{\mathrm{fail}}\ell^{2/3} + O(\ell \log(1/\ell)^4)$ (*read*, and conditional on C103, A4.1 and A4.2; independence credit 0; [R19]). These are statements about $d = 2$; Theorem 2.1 is a $d = 3$ statement.

6. Literature position

The closest located literature falls into four groups.

Persistent homology of random fields. Adler–Bobrowski–Borman–Subag–Weinberger [1] and Pranav [23] study persistent homology of random and Gaussian fields, principally through general structure and computation; Bobrowski–Borman [6] compute expected barcode descriptors of Gaussian fields (alternating sums of bar lengths over homological degrees), which integrate lifetimes rather than resolving the H_0 lifetime measure near zero. Feldbrugge, van Engelen, van de Weygaert, Pranav and Vegter [12] express expected two-dimensional superlevel persistence diagrams through candidate maximum–saddle densities and the probability that the saddle kills the component born at the given maximum: this is the selection-kernel architecture of §3, without a short-gap asymptotic. Chazal–Divol [10] prove absolute continuity of expected persistence diagrams for random point-cloud filtrations and treat Brownian sublevel sets; their results justify the existence of such densities in those settings and do not assert a general near-diagonal rate for smooth stationary Gaussian fields.

Critical points of Gaussian fields. Klein–Agam [16] describe the short-distance correlation of critical points, of order r^{2-d}, and Azaïs–Delmas [4] prove it, with explicit constants, for critical points of adjacent indices; in polar coordinates this is the $r\,dr$ pair measure behind (3.10). Ladgham–Lachièze-Rey [17] prove local repulsion of planar critical points, and Ancona–Gass–Letendre–Stecconi [3] analyze Kac–Rice singularities and cumulants. None of these imposes a height gap or the elder pairing, and the cubic fold exponent itself is classical.

Limit theorems for topological functionals and short features. Hirsch–Lachièze-Rey [14] prove functional central limit theorems for topological functionals of Gaussian fields on growing domains. Chenavier–Hirsch [11] prove Poisson limits for persistent cycles of random point clouds with extremal (minimal and maximal) lifetimes, and Hirsch–Lundbye–Otto [15] for large-lifetime cycles, including on the torus. Perez [22] obtains small-cutoff asymptotics of bar counts for continuous semimartingales, governed by quadratic variation; that concerns rough paths and counts above a cutoff, a different regularity class. Baryshnikov [5] computes the intensity of the H_0 persistence diagram of Brownian motion with drift m in closed form; near the diagonal it grows like $1/(m\Delta^3)$ in the lifetime Δ. This is the one-dimensional, rough-path counterpart of the near-diagonal question studied here. That paper says generalizing to higher-dimensional domains seems difficult but not entirely out of reach, at least for some classes of Gaussian processes, and will be pursued elsewhere.

Universality of persistence ratios. For random geometric complexes, Bobrowski and Skraba conjectured that suitably centered distributions of log log of the death/birth ratio are universal [7], and proved a weak form at fixed dimension and positive homological degree [8]. Heo [13], in a preprint, computes a second-order near-diagonal coefficient of the ratio law for degree-one Čech persistence in dimensions two and three, finds different values, and concludes that the strong form fails. These results concern point-cloud filtrations, positive homological degree and a multiplicative statistic; the present law concerns the additive lifetime of H_0 bars of a smooth Gaussian field.

Dimension one. Rice [24] and Cartwright–Longuet-Higgins [9] studied the heights of the maxima of a stationary Gaussian process, the latter also crest-to-trough heights for narrow spectra. The amplitude between a local maximum and the next minimum was studied by Lindgren [19], using Slepian-model conditional formulas, and by Lindgren–Rychlik [21]; Lindgren [18] shows that near a maximum of very low height the process is a random quartic, a low-level regime rather than a small-amplitude law. The exact joint law of adjacent extremes [20] contains the small-amplitude behaviour of §5.4 implicitly, but we have not found the $-1/3$ law stated there.

We have not located, in these sources or in the indexed and web searches recorded in [R21] (the latest on 13 October 2026), a prior theorem giving the fixed-side three-dimensional near-diagonal H_0 density (2.1), the coefficient reduction (2.9) or the correction (2.10), nor any near-diagonal lifetime law for the H_0 persistence of smooth Gaussian fields in dimension at least two. Substantial constituent mechanisms — the fold, the close-pair correlation, the selection kernel — are prior art. This is evidence of novelty, not a substitute for a systematic MathSciNet and zbMATH search with backward and forward citation chasing, which remains to be done before submission.

Relation to the parent record. The leading law (2.1) and the formula (2.9) are the case $d = 3$, $L = 24$ of Theorem C and (15.2) of the parent record [R2], which holds for every $d \geq 2$ and $L > 0$; the selection bound (3.13) is its Theorem A, and Proposition 3.10 is its (14.1), whose proof applies at every separation. Specific to the side-24 field are the closed form (2.11), the correction (2.10) with its error bound, the certified constants of Theorem 2.3(iii), and the second, three-dimensional route to (3.13) in Supplement §S5, whose finite-scale witness densities Supplement §S5.8 supplies (§3). The parent record is not a publication. Supplement Part III (Sections S12–S13) states its results with their proofs, in the reconciled reading given in [R2], among them the cap theorem, the marked Kac–Rice identity, the far bound (14.1) and the coefficient (15.2); where the record only sketches a step, the step is given as sketched there or written out in full. The other sections of the Supplement cite those statements, so the Supplement does not rely on the record for any statement it uses. Section 3 cites these Supplement statements and not the record: Supplement Theorem S12.3 for (3.13), and Supplement Proposition S6.2 for Proposition 3.10, whose proof uses from Part III only the marked Kac–Rice formula (Supplement Proposition S12.22). Supplement Theorem S5.36 gives a second, three-dimensional proof of (3.13).

7. Verification record and disclosure

Human review. No independent human specialist has reviewed this work.

AI review. All nonauthor checks of the proof chain to date were AI checks, carried out by models of Anthropic, OpenAI and xAI and by Grok Bot support agents (Cursor-hosted, distinct from the xAI model lane), posting from a single repository account under the author's direction. Every review record cites the exact file bytes it reviewed. No organizational independence is claimed for any review. In that sense: the leading term of Theorem 2.1 rests on the July/August 2026 proof (§3 and the Supplement [R1]) and on a parent chain with provider-distinct AI reviews [R2], whose Theorem A is the selection bound (3.13) (its second route in Supplement §S5, with Lemma S4.29 and Proposition S8.13 of the Supplement, has one nonauthor read by a Grok Bot support agent; a provider-distinct read by OpenAI found a gap in that route's finite-scale witness densities, Supplement draft 9 repairs it, and the repair has nonauthor reads by OpenAI (PASS_SCOPED, bound to the TEX source) and by a Grok Bot support agent (PASS); the other slices of that section have provider-distinct reads by xAI (PASS_SCOPED)); the rate in (2.1) and the bound (2.10) each have nonauthor reviews from two providers [R3, R4]; the lower bound (3.13a) has one provider's nonauthor review [R6]; the records behind Remark 3.7 have reads from two providers [R10], as does Remark 3.11 at its contact-integral scope [R20]; the links of Theorem 2.3 in [R11], and the certificates of c_1 and of the closed form of c_2, each have an OpenAI/Codex nonauthor review and an additional nonauthor read by Grok Bot agents; the finite-torus transfer of c_2 has an OpenAI `ACCEPT_SCOPED` of slices A–C and an Anthropic nonauthor acceptance of slices A and C, and no Grok Bot read, [R11, R12]; the results of Remark 2.4, and the $c_3\ell^{2/3}$ term and the rate δ in Remark 2.5, were read in bounded nonauthor slices [R14]; Theorems U, K and P and Theorem Z in that remark rest on records with one or two providers' nonauthor reviews [R7, R13]; and $B_{3,24} \approx 10.2$ is exploration only [R8]; and the Discussion labels its own items.

Computer assistance. The enclosures in Theorem 2.3(iii) and Remark 2.4(c) use outward-rounded interval arithmetic in the Python standard library, with replayable controls and rejected mutants in

the cited records. The exact triple-contact elimination (Supplement Lemma S5.15), the closed form of D_t in (3.24) and the margins of Theorem 3.6 were computed in exact arithmetic with SymPy and Python fractions; the elimination was re-checked independently in exact standard-library arithmetic (Remark 3.8), and $c_{3,24}$ is enclosed independently, to twenty digits [R5].

Remaining before submission. The MathSciNet/zbMATH pass of §6; completion of the Supplement [R1], which assembles the technical appendix and the records of Theorem 2.3 as numbered statements with proofs, including a reading of it as a whole (each of its sections has had a nonauthor reading) and the removal of provenance labels; and a journal-specific notation, bibliography and style pass. A finite-ℓ error band for (2.1) is not available (§4).

Appendix A. Repository records

The supplementary records cited as [Rn] are the following. Paths are in the repository `https://github.com/d6g8k5htny-coder/Math-` at the head of its `main` branch on 13 October 2026 (commit `b68d88b3`); "PR n" is the pull request that landed the record, with its merge commit, and hexadecimal prefixes after a path are Git blob identifiers. Review records are linked from each pull request.

[R1] *Supplement.* SIDE24 V3 supplement, checkpoint draft 13 (20 October 2026): `docs/manuscript/side24_v3_supplement_draft_20261013/` at commit `d28e6ab4` of the branch `claude/side24-v3-supplement-draft13-20261020` in `https://github.com/d6g8k5htny-coder/main`, cited as "Supplement" with its numbers S1.1, ... Its Part III (Sections S12–S13) transcribes [R2] in its reconciled reading. Parts I and II assemble, with the corrections recorded in its concordance, the SIDE24 technical appendix, V2 edition (Modules B–K, 1 August 2026), and `SIDE24_GAP_FILL_SUPPLEMENT.md` (Parts A–E, 1 August 2026; 48,972 bytes), both in the author's pre-review package. Independent exact re-elimination of the display (B.1) of 1 August: `reviews_side24_b1_display_claude_20260930/` (author's archive).

[R2] *Parent theorem.* `imports/lifetime_parent_20260925/UNIFORM_MATRIX_CAP_AND_LIFETIME.md` (`dfed3b8d`); Theorems A and C, (14.1), (15.2); with its cap proof `imports/lifetime_parent_20260925/MARKED_CYLINDER_CAP_PROOF.md` (`0633aca3`). It is read as reconciled in `reviews/d1_chain_reconciliation_20260928/RECONCILIATION.md` (`75da2597`): with the congruence erratum `imports/lifetime_parent_20260925/ERRATUM_CONGRUENCE.md` (`213594d6`), with its Section 9 replaced by `reviews/d1_section9_borel_repair_20260925/REPAIR.md` (`fe9b9ce4`), and with the wording W1 and the embedding radius of that record's §5. Full-depth nonauthor review of D1-C/D/E: PR 143 (`3216e2e7`).

[R3] *Theorem R.* `frontiers/three_fronts_20260924/LIFETIME_REMAINDER.md` (`247b3ecf`). Reviews: xAI, slices R1–R6 (`main` issue 67); Anthropic, full depth, PR 186 (`c839f286`).

[R4] *The correction (2.10).* `coefficients/side24_remainder_20260929/PROOF.md` (`0bcf0a6f`; PR 154, `8e61fc4c`); `frontiers/side24_periodization_remainder_20260929/PROOF.md` (`03291568`; PR 156, `8ba851fb`); first variation: `frontiers/side24_periodization_20260929/PROOF.md` (`1a84f215`; PR 144, `0cd0b821`).

[R5] *Coefficient enclosure.* `coefficients/side24_v1` (a twenty-digit enclosure of $c_{3,24}$, and the periodization bound $|c_{d,24}/c_{d,\mathrm{ref}} - 1| < 10^{-106}$ for $d = 2, 3$), with the independent recomputation `reviews/side24_v1_coefficient_claude_20260929`.

[R6] *Lower bound (3.13a).* `frontiers/window_multiplicity_laws_20260928/ELDER_LOWER_AND_DENSITY_GAP.md` (`aaefc8da`); `frontiers/elder_dimension_lift_20260928/PROOF.md` (`7303bd79`); `frontiers/elder_lower_all_d_20260929/PROOF.md` (`7f41c9e3`; PR 149, `21244e5e`; Theorem E1$_d$).

[R7] *Unrestricted marks.* Theorem U: `frontiers/unrestricted_selection_difference_20260929/PROOF.md` (`5a55b179`); Theorem K and Corollary T: `frontiers/c7_total_bounded_20260929/PROOF.md` (`28748b08`; PR 150, `7f438fca`), with `ERRATUM_NORMALIZED_FORM.md` (PR 289, `5b012335`); Theorem Z: `frontiers/c7_zero_gap_limit_20260929/PROOF.md` (`5b6328ea`; PR 155, `7a1cb09a`). The Supplement proves the bound (K2) of Theorem K as its Lemma S8.16 (draft 11)

and Theorem Z as its Proposition S8.4(i)–(ii), in its §S8.6 (draft 12); with these, Part II of the Supplement imports no analytic statement from a repository record, and its only computer-assisted inputs are the certified enclosures of c_1 and c_2 (Supplement §§S9, S11).

[R8] *Equal-height mass.* `frontiers/equal_height_mass_value_20261001/` (PR 211, `94f9e75e`): Proposition V (volume law); γ_3 and $B_{3,24}$ are exploration.

[R9] *Far elder pairs.* `frontiers/far_elder_rate_20260930` (PR 187, `c2f1270d`; rate $O(\ell^{2/3})$); `frontiers/far_elder_flat_ridge_20260930` (PR 188, `f7f083ec`; Theorem G, rate $O(\ell^N)$, with the moving-cutoff extension C89).

[R10] *Rejection mechanism and coefficient.* `frontiers/local_elder_geometry_20260930` and companions (PR 170, `fa2e9909`; Theorems E and S); `frontiers/concave_fibre_elder_20260930` (PR 175, `eb659bd3`; Theorem F); `frontiers/local_pairing_20260930` (PR 180, `4522be59`); `frontiers/soft_fold_limit_20261002` (PR 243, `fa8b0d21`).

[R11] *The three-term law.* Theorem R^+: `frontiers/remainder_vanishing_20260930` (PR 191, `95565474`; errata PR 270, `ced1fca9`); Theorem R^{++}: `frontiers/remainder_rate_20260930` (PR 198, `124c37d9`); Theorems CU and CU′: `frontiers/cusp_second_order_20261001` (PR 207, `566b1a14`); Theorem T: `frontiers/candidate_third_order_20261001` (PR 218, `fb6ee97b`); Theorem E3: `frontiers/elder_third_order_20261001` (PR 220, `0d797780`); Theorems T^+ and $E3^+$: `frontiers/third_order_rate_20261001` (PR 229, `6f74f7a7`); the third coefficient: `frontiers/third_order_coefficient_20261001` (PR 216, `b718f235`).

[R12] *Certified constants.* c_1: `frontiers/cusp_coefficient_certified_20261001` (PR 219, `5a836a8e`), with its erratum `ERRATUM_20261013.md` (`a9c3f95a`; PR 454, `a6116d8a`, landed after `b68d88b3`): the enclosure is of the ratio of the integral (2.13) to $c_{3,24}$, and its reading as a coefficient ratio uses Theorem CU, as Theorem 2.3(iii) states; c_2 in closed form: `frontiers/c2_exact_20261001` (PR 223, `f9ebba17`); torus transfer of c_2: `frontiers/c2_finite_jet_transfer_20261001` (PR 232, `7fe06b01`).

[R13] *Theorem P.* `frontiers/candidate_parity_rate_20261001` (PR 237, `cb73b117`).

[R14] *The order $\ell^{2/3}$.* Notes TL: `frontiers/tl_soft_layer_20261006` (PR 384, `f3318db6`); TS: `frontiers/ts_small_kappa_20261006` (PR 387, `40440ed8`); EM: `frontiers/em_intermediate_elder_20261006` (PR 388, `ebbf3b7f`); EM.1, LU, C3 and SL: `frontiers/lifetime_two_thirds_chain_20261007` (PR 390, `7d2f6250`); C7: `frontiers/c7_elder_two_thirds_20261007` (PR 392, `7def48d4`); V24 and DC: `frontiers/v24_dc_two_thirds_sign_20261007` (PR 393, `73734950`); CJ: `frontiers/lifetime_cj_model_j_20261012` (PR 442, `ea2f3e25`); DD: `frontiers/lifetime_dd_model_d_20261013` (PR 450, `a115feeb`); C3-R: `frontiers/lifetime_c3r_rate_20261013` (PR 452, `dbe6c0a1`).

[R15] *Soft rejected pairs.* `frontiers/soft_rejected_pairs_20261002` (PR 242, `54c6759e`); the soft model in closed form, Theorems A and B: `frontiers/soft_closed_form_20261002` (PR 244, `07320089`), conditional on PR 170 Theorem E(1) and its cubic-model input.

[R16] *Simulation.* The full-field Monte Carlo records of PR 216 (`b718f235`) and the four-term comparison recorded with note V24.

[R17] *Dimension one.* Theorem 1D: `frontiers/d1_third_order_law_20261001` (PR 214, `e4ca2b3f`); Theorem $1D^+$: `frontiers/d1_sharp_remainder_20261001` (PR 238, `8404169d`); on the line: `frontiers/d1_line_20261004` (PR 271, `4242ab95`, with its wording successor PR 286).

[R18] *Dimension three, compact marks.* `frontiers/qs_d3_soft_layer_chain_20261005` (PR 359, `298b2631`), storing the addenda A3–A3.11 and their reads; the hard-fibre reduction is A3 and A3.1 with their erratum, and the rate rests on the author-side chain A3.3–A3.8.

[R19] *The plane.* `frontiers/planar_soft_layer_chain_20261003` (PR 249, `e0bed3ed`); `frontiers/planar_compact_k_chain_20261006` (PR 360, `930bd976`).

[R20] *Torus contact coefficients.* Record C104 (`main` issue 229, comment 5967953399) and its reviews (comments 5968193256 and 5971201678, the latter with the constants evaluated).

[R21] *Literature searches.* `reviews/literature_lifetime_law_recon_20260929/RECONNAISSANCE.md` (`003b9879`; PR 147); `reviews/literature_classical_neighbors_20261001` (PR 210, `0cdc19fc`),

including an indexed-database pass of 4 October 2026; two independent web searches of 13 October 2026 (`main` issue 275, comment 6146291432).

References

1. Robert J. Adler, Omer Bobrowski, Matthew S. Borman, Eliran Subag, and Shmuel Weinberger, *Persistent homology for random fields and complexes*, Borrowing Strength: Theory Powering Applications—A Festschrift for Lawrence D. Brown, IMS Collections, vol. 6, Institute of Mathematical Statistics, Beachwood, OH, 2010, pp. 124–143.
2. Robert J. Adler and Jonathan E. Taylor, *Random fields and geometry*, Springer Monographs in Mathematics, Springer, New York, 2007.
3. Michele Ancona, Louis Gass, Thomas Letendre, and Michele Stecconi, *Zeros and critical points of Gaussian fields: cumulants asymptotics and limit theorems*, 2025, arXiv:2501.10226.
4. Jean-Marc Azaïs and Céline Delmas, *Mean number and correlation function of critical points of isotropic Gaussian fields and some results on GOE random matrices*, Stochastic Process. Appl. **150** (2022), 411–445, arXiv:1911.02300.
5. Yuliy Baryshnikov, *Brownian motions, persistent homology and chirality*, J. Appl. Comput. Topol. **9** (2025), Paper No. 28.
6. Omer Bobrowski and Matthew Strom Borman, *Euler integration of Gaussian random fields and persistent homology*, J. Topol. Anal. **4** (2012), no. 1, 49–70, arXiv:1003.5175.
7. Omer Bobrowski and Primoz Skraba, *A universal null-distribution for topological data analysis*, Sci. Rep. **13** (2023), 12274, arXiv:2207.03926.
8. ———, *Universality in random persistent homology and scale-invariant functionals*, 2024, arXiv:2406.05553.
9. D. E. Cartwright and M. S. Longuet-Higgins, *The statistical distribution of the maxima of a random function*, Proc. Roy. Soc. London Ser. A **237** (1956), 212–232.
10. Frédéric Chazal and Vincent Divol, *The density of expected persistence diagrams and its kernel based estimation*, J. Comput. Geom. **10** (2019), no. 2, 127–153, arXiv:1802.10457.
11. Nicolas Chenavier and Christian Hirsch, *Extremal lifetimes of persistent cycles*, Extremes **25** (2022), no. 2, 299–330.
12. Job Feldbrugge, Matti van Engelen, Rien van de Weygaert, Pratyush Pranav, and Gert Vegter, *Stochastic homology of Gaussian vs. non-Gaussian random fields: graphs towards Betti numbers and persistence diagrams*, J. Cosmol. Astropart. Phys. **2019** (2019), no. 9, 052, arXiv:1908.01619.
13. Eunwoo Heo, *Near-diagonal asymptotics and the failure of strong universality in random čech persistence*, 2026, arXiv:2610.08257.
14. Christian Hirsch and Raphaël Lachièze-Rey, *Functional central limit theorem for topological functionals of Gaussian critical points*, Ann. Appl. Probab. **36** (2026), no. 5, 4430–4464, arXiv:2411.11429.
15. Christian Hirsch, Nikolaj Nyvold Lundbye, and Moritz Otto, *Poisson approximation of large-lifetime cycles*, Extremes **29** (2026), no. 3, 495–559, arXiv:2412.17482.
16. Avraham Klein and Oded Agam, *Critical point correlations in random Gaussian fields*, J. Phys. A **45** (2012), no. 2, 025001.
17. Safa Ladgham and Raphaël Lachièze-Rey, *Local repulsion of planar Gaussian critical points*, Stochastic Process. Appl. **166** (2023), Paper No. 104221, arXiv:2209.04150.
18. Georg Lindgren, *Some properties of a normal process near a local maximum*, Ann. Math. Statist. **41** (1970), 1870–1883.
19. ———, *Wave-length and amplitude in Gaussian noise*, Adv. in Appl. Probab. **4** (1972), 81–108.
20. ———, *Gaussian integrals and Rice series in crossing distributions—to compute the distribution of maxima and other features of Gaussian processes*, Statist. Sci. **34** (2019), no. 1, 100–128.
21. Georg Lindgren and Igor Rychlik, *Wave characteristic distributions for Gaussian waves—wave-length, amplitude and steepness*, Ocean Engineering **9** (1982), no. 5, 411–432.
22. Daniel Perez, *On the persistent homology of almost surely C^0 stochastic processes*, J. Appl. Comput. Topol. **7** (2023), no. 4, 879–906, arXiv:2012.09459.
23. Pratyush Pranav, *Topology and geometry of Gaussian random fields II: on critical points, excursion sets, and persistent homology*, 2021, arXiv:2109.08721.
24. S. O. Rice, *Mathematical analysis of random noise*, Bell System Tech. J. **23** (1944), 282–332, and 24 (1945), 46–156.